4.1 - Newton's Method

Goals: Apply Newton's method to estimate solutions to equations
Before Class Videos: No video; read Section 4.1 of APEX Calculus 1.
Note: We will use the following DESMOS extensively: `https://www.desmos.com/calculator/y1gjz9bjwn`

Example 1: Recall how IVT implies $f(x) = x^3 - x - 1$ has a root in $(0, 2)$.

Can I use my knowledge of what happens around $x = 1$ to estimate where the root is?

Newton's Method: Goal: $f(x) = 0$. Method: Starting at x_0 estimate a new location x_1 such that $f(x_0) + f'(x_0)(x_1 - x_0) = 0$. We continue iteratively via

$$\boxed{x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}}.$$

Example 2: $x^4 - 2 = 0$ and start with $x_0 = 1$.

Example 3:

$$f(x) = \begin{cases} -\sqrt{1-x}, & x < 1, \\ \sqrt{x-1}, & x \geq 1. \end{cases}$$

With $x_0 = 2$,

Example 4: $\sqrt[3]{x} = 0$ starting at $x_0 = 1$.

Example 5: Estimate where $\cos(x) = 0$. Start $x_0 = 1$.

Suggested after class problems from APEX: Section 4.1: 3, 9, 15